Colour of Coordination Complexes : The colour of complexes originates from two sources : (1) *d-d* transition and (2) charge transfer transition.

A complex absorbs the required energy for a particular transition from the surrounding white light and obviously transmits the remaining complementary colour. This transmitted complementary colour is the colour that we observe (see section 23.2.5 Part II).

d-d transition : In its ground state the d^1 electron of $[Ti(H_2O)_6]^{3+}$ is in the t_{2g} set. On irradiation with blue-green light of energy equal to the Δ (493 mμ = 20,300 cm^{-1} = 20.3 *kK*) of the hexaaquatitanium(III) ion, the complex will absorb such energy to allow a transition of the t_{2g} electron to the excited e_g set. When we see the complex through transmitted light it appears purple violet. For multielectron system interactions between *d*-electrons gives rise to many electronic states so that more than one transition is possible (Chapter 11). Quantitative expressions have been derived for spectral transitions on the crystal field model connecting energy of the transition, crystal field strength and interelectronic repulsion. Furthermore quantitative expressions are also available connecting magnetic moments with 10 *Dq*, which connections were not forthcoming from VB theory.

It can be guessed that if H_2O of $[Ti(H_2O)_6]^{3+}$ is substituted by a stronger ligand, say by bipyridine giving $[Ti(bpy)_3]^{3+}$, then light of much higher energy (presumably of the violet region ; ~ λ = 400 nm ; ~ 25,000 cm^{-1} ; ~ 25 kK] will be required to effect the $t_{2g} \rightarrow e_g$ electronic transition since Δ will now be much higher. The transmitted light will be correspondingly of lower energy and thus the observed colour will be yellow-green or so. Thus we have a simple crystal field explanation of changing colour of a complex with change in the coordinating ligand, assuming however the geometry remains the same. This is, in fact, how the spectrochemical series is obtained.

It is also of interest to note that high-spin cobalt(III) ($3d^6$) fluoro complex $K_3[CoF_6]$ looks blue while all low-spin octahedral complexes of cobalt(III) (eg : $[Co(NH_3)_6]Cl_3$) are yellow to orange. Evidently $K_3[CoF_6]$ absorbs low energy light and transmits high energy light while the diamagnetic complex does the reverse i.e. absorbs high energy light and transmits low energy light. A simple undergraduate level explanation is that for the high-spin complex Δ is less than the pairing energy *P* of cobalt(III) while for the low-spin complex Δ is much larger than *P* i.e. Δ(l.s.) >> Δ(h.s.): Hence different amounts of energy are required for the electronic transitions in the two spin states, thereby showing varying colours. However this explanation is too simple. For an exact explanation one has to make use of the Tanabe-Sugano diagrams of h.s. d^6 (Fig. 11.10) and l.s.d^6 (Fig. 11.11).

Charge transfer transition : There are many compounds where either the metal ion is highly oxidising and the ligands are reducing or that the metal ion is highly reducing and the ligands are oxidising. In such cases there occurs transfer of charge (i.e. electron) from the reducing partner to the oxidising partner. The metal ions may or may not possess *d*-electrons. Common examples are intense purple permanganate (Mn(VII), $3d^0$), brick-red mercury(II) iodide (Hg(II), $5d^{10}$), yellow chromate (Cr(VI), $3d^0$), dark red $[Fe(bpy)_3]^{2+}$ (Fe(II), $3d^6$), blood red iron(III) thiocyanate (Fe(III),$3d^5$), etc.

The selection rules of electronic transitions are :

1. No electronic transition is permitted where the ground state and the excited state have different spin arrangements i.e. have different spin multiplicities. Only those transitions are

allowed for which $\Delta S = 0$ i.e. *transitions are permitted between states of the same spin multiplicity.*

2. *Only those electronic transitions are allowed for which* $\Delta l = \pm 1$. This selection rule due to Laporte points out that no electronic transitions are allowed for which $\Delta l = 0$ or ± 2. Thus transitions such as $2s \rightarrow 2p$, $2s \rightarrow 3p$, $3p \rightarrow 3d$ are allowed but $3d \rightarrow 3d$, $1s \rightarrow 2s$ or $3s \rightarrow 3d$ are not. No electronic transition can occur merely because of redistribution of electrons in similar orbitals i.e. *d-d* transitions (such as $t_{2g} \rightarrow e_g$ transition in $[Ti(H_2O)_6]^{3+}$) are forbidden. For quantum mechanical proof of the two selection rules see Appendix (Part II).

Although *d-d* transitions are forbidden yet the coordination complexes of transition metals are coloured. This indicates that Laporte selection rule must be relaxed in practice. Whenever a complex lacks a centre of symmetry (e.g. : in a tetrahedron) or that odd vibrational effects lead to a change of bond lengths or bond angles there is scope for mixing of metal 3*d*-orbitals with metal 4*p* orbitals or with ligand *p*-orbitals. The pure *d*-character of the transition metal *d*-orbitals is then lost and Laporte selection rule is relaxed. A situation arises so that electron transfer from a *d*-orbital to a *p*-orbital can occur. Forbidden transitions have low molar extinction coeficients (10 to 200 l mol^{-1} cm^{-1}) for six-coordinate complexes and (100—1000 l mol^{-1} cm^{-1}) for tetrahedral complexes. In a non-centrosymmetric tetrahedral complex the selection rules are relaxed to a greater extent.

Charge transfer transitions may be either from ligand to metal ($L \rightarrow M$) or from metal to ligand ($M \rightarrow L$). In such complexes the ligand *p*-orbitals and metal *d*-orbitals interact to form M.O.'s. When these M.O.'s are arranged in order of their energy it is found that some M.O.'s have predominantly ligand (i.e. *p*) character while some M.O.'s retain predominantly metal (i.e. *d*) character. Laporte selection rule allows $p \rightarrow d$ transition ($\Delta l = +1$) so that charge transfer transitions are allowed transitions with high molar extinction coefficients (10^3 to 10^5 l mol^{-1} cm^{-1}). Such transitions are usually high energy transitions and occur in the ultraviolet region. Charge transfer complexes appear coloured to our eyes only when the transitions occur in the near visible region or when the transition in the u.v. has a long tail in the visible. Common examples of $L \rightarrow M$ charge transfer are mercury(II) iodide, MnO_4^-, CrO_4^{2-} etc. Examples of $M \rightarrow L$ transition include $[Fe(o\text{-}phen)_3]^{2+}$, $[Fe(bpy)_3]^{2+}$ etc. While *d-d* transitions can occur only in complexes of transition metal ions with incomplete *d*-level, charge transfer can occur in both transition metals with incomplete *d*-level as also in d^0, d^{10} and non-transition metal compounds. Charge transfer transitions are also known as *redox transitions.*

The more reducing M is and the more oxidising L is, the lower is the energy of the $M \rightarrow L$ charge transfer transition. The absorption maximum of $[Fe(bpy)_3]^{2+}$ occurs at 515 mμ while for $[Ni(bpy)_3]^{2+}$, $[Co(bpy)_3]^{2+}$ the maximum is at still lower wavelength range. Iron can be estimated down to 10^{-6} g/ml via formation of $[Fe(bpy/o\text{-}phen)_3]^{2+}$. The more oxidising M is and the more reducing L is the lower is the energy of the $L \rightarrow M$ charge transfer transition. Thus iron(III) complexes give more $L \rightarrow M$ charge transfer transition than does chromium(III). Most colorimetric analyses of trace quantities of metal ions utilise charge transfer transitions. *d-d* transitions, being of low intensity, are of little value for such purposes.

Charge transfer spectra need not be restricted to $M \rightarrow L$ or $L \rightarrow M$ type. A third less common type is exhibited by the so-called Prussian blue or Turnbull's blue precipitate $KFeFe(CN)_6$. One of the two irons is Fe(III) while the other is Fe(II). Electron transfer is possible from Fe(II) to Fe(III). Note that $K_2FeFe\ (CN)_6$ is colourless because both the irons are Fe(II) and thus there exists no scope for any (redox) charge transfer transition. Given below are the salient distinctions between *d-d* transitions and charge transfer transitions.

d-d transitions	*charge transfer transitions*
1. Presence of *d* electrons essential.	1. Presence of *d* electron not essential. It may occur in d^0, d^n, d^{10} systems.
2. Laporte forbidden.	2. Laporte allowed.
3. Low molar extinction coefficient (5–200 l mol^{-1} cm^{-1}).	3. Extremely high molar extinction coefficient (5,000 ~ 35,000 l mol^{-1} cm^{-1}).
4. No redox system involved.	4. Redox system essential.
5. Not useful for spectrophotometric estimation of metal ions.	5. Very useful for spectrophotometric estimation of metal ions.

Intensities of forbidden and allowed transitions are given below.

type of transitions	ϵ_{max} (l mol^{-1} cm^{-1})
spin forbidden	~ 1
Laporte forbidden $d-d$	10 – 200 (octahedral complexes)
Laporte (partially) allowed $d-d$	100 – 1000 (tetrahedral complexes)
charge transfer (Laporte allowed)	5,000 – 35,000

More on the electronic spectra from the viewpoint of the crystal field theory is given in Chapter 11.

Passage of Crystal Fields to Ligand Fields : The crystal field model does not consider any covalent interaction between ligands and the metal ion. Experience shows that in several instances overlap of orbitals of the metal ion and the ligands does occur.

(1) In a low-spin hexachloroiridate(IV), $[IrCl_6]^{2-}$ (t_{2g}^5) the single unpaired electron has been shown from an analysis of the electron spin resonance spectrum to be delocalised into the six chloride ions. The multiband ESR spectrum convincingly proves that the iridium unpaired electron is definitely disturbed by the nuclei of the chloride ligands. This is not possible unless there occurs an overlap of metal—ligand orbitals. For details on the ESR evidence see Appendix V, Part II.

(2) Intensities of the forbidden *d-d* transition cannot be adequately explained on the basis of mixing of metal *d* and metal *p/s* orbitals alone or on the basis of vibrations of the complex but that metal *d*-orbitals have to be assumed to be mixed up with ligand *p*-orbitals also.

(3) Crystal field spectral bands cannot be fitted precisely if the interelectronic repulsion of metal electrons in a complex be assumed to be the same as in the free ion (section 11.10.5). Instead, good fits are obtained when the interelectronic repulsion is assumed to be 20-25% less in a complex than in a free ion. It follows that the *d*-orbitals have expanded in size and have penetrated into ligand orbitals so that repulsions between metal *d*-orbitals becomes smaller than in the free ion. The ligand electrons partially screen the *d*-orbital electrons from the nuclear charge of the metal ion. Thus the *d*-orbital electrons spend sometime on the ligands